Machine Learning Project using Neural Network

Report PDF

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1 Introduction

- This project implements an Artificial Neural Network (ANN) to solve a classification problem using the Credit Card Customer Churn dataset obtained from Kaggle.

- The objective of this project is to preprocess the data, perform exploratory data analy- sis (EDA), train an ANN model, and evaluate its performance using different classification metrics.

- The model predicts whether a customer will leave the bank or remain with the bank.

2 Dataset Description

The dataset contains 10,000 customer records with several input features and one target variable.

The target variable is:

- 0 = Customer stayed
- 1 = Customer left

The dataset includes customer information such as:

- Credit Score
- Geography
- Gender
- Age
- Balance
- Number of Products
- Active Member Status
- Estimated Salary

- The objective is to classify customers into churn and non-churn categories.

3 Methodology

3.1 Exploratory Data Analysis (EDA)

The dataset was analyzed using functions such as head(), shape, and info() to understand the structure of the data.

- Missing values were checked and no major missing values were found. - A count plot was used to visualize customer churn distribution.
- A correlation heatmap was used to identify relationships among features.

3.2 Data Preprocessing

The dataset was preprocessed before training.

The following steps were performed:

- Removed unnecessary columns
- Encoded categorical variables
- Standardized numerical features
- Split data into training and testing sets

The dataset was divided into:

- 80% training data
- 20% testing data

- Feature scaling was applied to improve model performance.

3.3 ANN Model Development

An ANN model was built using Keras.

The model contains:

- Input layer
- First hidden layer with 16 neurons
- Second hidden layer with 8 neurons
- Output layer with 1 neuron

Activation functions used:

- ReLU for hidden layers
- Sigmoid for output layer

The model was compiled using:

- Optimizer: Adam
- Loss function: Binary Crossentropy

Evaluation metrics included:

- Accuracy
- Precision
- Recall
- F1-score

3.4 Model Training

The model was trained for:

- 20 epochs
- Batch size = 32

- Validation data was used during training to monitor performance.

4 Results

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

- The ANN model achieved an accuracy of **85.6%** on the testing data.

- Out of 2000 testing samples, the model correctly predicted **1712 customers** and incorrectly predicted **288 customers**.

- The confusion matrix showed that the model classified most customers correctly, while a smaller number of customers were misclassified.

- The training and validation accuracy curves remained close to each other, which indicates that the model did not show significant overfitting.

5 Conclusion

- In this project, an Artificial Neural Network was successfully implemented for customer churn prediction.

- The model showed good performance in classifying whether a customer will leave the bank or stay.

- This project demonstrated that ANN can be used effectively for customer churn prediction in the banking sector.

Final Note:

The current model gives good accuracy, but it can be improved by using more hidden neurons, training for more epochs, and adding dropout to reduce overfitting.